

Lightweight HyDE-Based Semantic Alignment for Personalized Educational Recommendation

1st Given Name Surname
dept. name of organization (of Aff.)
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City, Country
email address or ORCID

2nd Given Name Surname
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email address or ORCID

3rd Given Name Surname
dept. name of organization (of Aff.)
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email address or ORCID

4th Given Name Surname
dept. name of organization (of Aff.)
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City, Country
email address or ORCID

5th Given Name Surname
dept. name of organization (of Aff.)
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City, Country
email address or ORCID

6th Given Name Surname
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Abstract—This document is a model and instructions for L^AT_EX. This and the IEEEtran.cls file define the components of your paper [title, text, heads, etc.]. ***CRITICAL: Do Not Use Symbols, Special Characters, Footnotes, or Math in Paper Title or Abstract.**

Index Terms—component, formatting, style, styling, insert.

I. INTRODUCTION

The growing availability of digital learning resources, training activities, internships, and professional-development opportunities has increased the need for effective personalization in higher education. Personalized recommender systems support learners in navigating heterogeneous educational content and opportunities by considering interests, competencies, learning needs, and goals [1]. Recent approaches increasingly rely on semantic representations and dense text embeddings, which have been applied to learning resource recommendation, academic decision support, and student–internship matching [2], [3]. However, learner-side information and candidate educational resources are often expressed from different semantic perspectives: learner profiles encode interests, competencies, preferences, and intentions, whereas candidate items describe learning outcomes, requirements, activities, or expected competencies. Direct embedding-based matching therefore assumes that the representation model can implicitly bridge differences in terminology, abstraction, and discourse structure, although dense encoders do not explicitly align learner intent with the semantic form of the opportunities expected to satisfy it.

Large Language Models (LLMs) provide a potential mechanism for reducing this representation gap and have increasingly been used in recommender systems for preference interpretation, representation learning, and recommendation generation [4]. Rather than employing an LLM as the final recommendation engine, a more controlled approach is to use generation as an intermediate semantic transformation

step while retaining retrieval over a fixed corpus of real candidate items. Related information-retrieval methods include Query2doc [5], query rewriting [6], and Hypothetical Document Embeddings (HyDE) [7], which generate or reformulate representations prior to retrieval. HyDE, in particular, transforms an input query into a hypothetical target document and embeds this generated representation before retrieving semantically related real documents. However, these methods have primarily been formulated for query–document retrieval, whereas personalized educational recommendation requires matching learner characteristics, needs, and objectives with suitable opportunities rather than retrieving documents based on topical relevance alone.

This work investigates HyDE as a semantic alignment mechanism for personalized educational recommendation. Learner information is transformed by an instruction-following LLM into a hypothetical target-oriented representation, which is subsequently embedded and compared with a corpus of real educational resources or opportunities. In this formulation, the LLM performs semantic transformation while the final recommendation remains grounded in actual candidate items. The approach is evaluated across two complementary higher-education recommendation tasks: student–internship matching and personalized capacity-building resource recommendation. In addition, the study examines whether lightweight, locally deployable language models can retain the benefits of semantic alignment while reducing computational requirements in resource-constrained educational environments.

The main contributions are:

- A formulation of educational recommendation as a semantic alignment problem between heterogeneous learner-side and target-side representations.
- A HyDE-based framework that transforms learner information into a target-oriented representation prior to dense retrieval over real educational resources and opportunities.

- An evaluation across two higher-education recommendation tasks against lexical and direct embedding-based baselines.
- An analysis of lightweight local LLMs, focusing on the trade-off between recommendation effectiveness and computational requirements.

II. RELATED WORK

Personalized recommender systems have been extensively investigated in technology-enhanced and higher education to support the selection of learning resources, courses, academic pathways, and professional opportunities. Existing approaches include collaborative, content-based, knowledge-based, and hybrid recommendation strategies, with recent studies increasingly incorporating semantic representations of learner and item information [1], [8]. In particular, dense text representations have enabled semantic matching in settings where explicit interaction histories are limited. Goh et al. applied sentence embeddings and similarity scoring to student–internship recommendation [2], while Seetaworn et al. combined student competencies, course information, job descriptions, and knowledge graphs to generate personalized internship recommendations [3]. These approaches demonstrate the value of semantic information for educational recommendation, but primarily perform matching directly between representations derived from the learner and candidate items.

The emergence of Large Language Models (LLMs) has extended recommender systems beyond conventional representation learning by enabling richer interpretation of user preferences, textual side information, and contextual information [4]. Similar developments have appeared in education, where LLMs and retrieval-augmented generation have been used to provide personalized and grounded learning support. Retrieval augmentation has, for example, been used with lightweight language models to improve the reliability of educational coding assistance [9], while recent systems have combined retrieval and generative models to improve personalization and factual grounding. Such approaches primarily employ retrieval to support subsequent generation. In contrast, the present problem concerns the use of generation *before* retrieval to improve the representation used for recommendation.

Generative retrieval methods provide a relevant foundation for this direction. Query2doc uses LLM-generated pseudo-documents to enrich queries prior to sparse or dense retrieval [5], while Rewrite-Retrieve-Read reformulates queries to improve their alignment with the downstream retrieval process [6]. HyDE introduces a related zero-shot approach in which an instruction-following language model generates a hypothetical document whose embedding is used to retrieve nearby real documents [7]. These methods demonstrate the value of transforming a source representation toward the semantic form of its retrieval target. However, they have mainly been studied in conventional query–document retrieval rather than personalized educational recommendation, where learner profiles and candidate opportunities represent different entity types and recommendation relevance depends on compatibility rather

than topical relevance alone. This distinction motivates the investigation of HyDE-based semantic alignment for learner-to-opportunity and learner-to-resource recommendation, together with the use of lightweight language models for deployment in resource-constrained educational environments.

III. PROPOSED FRAMEWORK

The proposed framework addresses personalized educational recommendation as a semantic alignment problem between learner-side information and target-side educational opportunities or resources. Instead of directly comparing the original learner representation with candidate items, an intermediate LLM-generated representation is introduced to reduce the semantic asymmetry between the two sides of the recommendation task.

Let q denote a learner profile containing information such as interests, competencies, needs, and objectives, and let $\mathcal{D} = \{d_1, d_2, \dots, d_N\}$ denote the corpus of candidate educational items. Direct embedding-based recommendation represents both q and d_i using an embedding function $E(\cdot)$ and ranks candidates according to their cosine similarity:

$$s_{\text{direct}}(q, d_i) = \frac{E(q) \cdot E(d_i)}{\|E(q)\| \|E(d_i)\|}. \quad (1)$$

The proposed formulation introduces a generative transformation $G(\cdot)$ before retrieval. Following the principle of Hypothetical Document Embeddings (HyDE) [7], the learner representation is transformed into a hypothetical target-side description:

$$h_q = G(q), \quad (2)$$

where h_q represents an educational opportunity or resource expected to be compatible with the learner profile. Recommendation is subsequently performed by comparing the embedding of h_q with the pre-computed embeddings of the candidate corpus:

$$s_{\text{HyDE}}(q, d_i) = \frac{E(h_q) \cdot E(d_i)}{\|E(h_q)\| \|E(d_i)\|}. \quad (3)$$

Candidate items are ranked in descending order of $s_{\text{HyDE}}(q, d_i)$. The language model is therefore used only as an intermediate semantic transformation component; the final recommendations remain grounded in the predefined corpus and are not generated by the LLM.

The framework consists of three stages: (i) LLM-mediated generation of a target-oriented hypothetical representation, (ii) dense encoding of the generated representation and candidate items, and (iii) cosine-similarity-based ranking. Candidate embeddings are computed offline and reused across queries, limiting the online computational cost to a single LLM generation and one query embedding operation. This separation also enables the generative component to be replaced by language models of different capacities without modifying the retrieval architecture.

TABLE I
EVALUATION DATASET STATISTICS.

	Internship	Educational Resources
Profiles	XX	XX
Candidate items	XX	XX
Relevance judgments	XX	XX
Avg. profile length	XX	XX
Avg. item length	XX	XX

IV. METHODOLOGY

A. Educational Recommendation Tasks

The framework is evaluated on two complementary higher-education recommendation tasks. The first considers student–internship matching, where learner profiles are matched with internship or professional-placement opportunities. The second considers personalized educational resource recommendation, where learner or academic-staff needs are matched with digital learning and capacity-building resources.

The two tasks provide different recommendation settings while preserving the same underlying challenge: the source representation describes characteristics and intentions of an individual, whereas the target representation describes properties of an educational resource or opportunity. This enables evaluation of whether the proposed semantic alignment mechanism generalizes across different educational recommendation scenarios.

For the student–internship task, the evaluation set contains N_s learner profiles and N_i internship opportunities. For the educational-resource task, the evaluation set contains N_l learner or staff profiles and N_r candidate resources. Final dataset statistics are reported in Table I.

B. Retrieval Configurations

Four main retrieval configurations are considered.

BM25 provides a lexical retrieval baseline and measures performance without dense semantic representations.

Direct-E5 represents the learner profile and candidate items directly using the `intfloat/e5-small-v2` sentence-embedding model. Candidates are ranked using cosine similarity according to Eq. 1.

HyDE-Small applies the proposed semantic-alignment stage using a compact instruction-following language model before embedding the generated target-side representation with `e5-small-v2`.

HyDE-Medium follows the same procedure using a higher-capacity local language model, allowing the effect of model capacity on recommendation effectiveness to be examined.

An additional language model from a different model family is included to determine whether the observed behaviour is specific to a particular generative model architecture.

C. LLM-Based Semantic Alignment

The same instruction template is used across all generative models to ensure a controlled comparison. Given a learner-side description, the model is instructed to generate a concise hypothetical description of an educational opportunity or resource

TABLE II
EVALUATED RECOMMENDATION CONFIGURATIONS.

Method	Generator	Retrieval
BM25	–	Lexical
Direct-E5	–	E5 + cosine
HyDE-Small	Small local LLM	E5 + cosine
HyDE-Medium	Medium local LLM	E5 + cosine
HyDE-CrossModel	Alternative LLM	E5 + cosine

that would satisfy the learner’s characteristics and objectives. No candidate information is provided during this stage.

Generation parameters, including maximum output length and decoding settings, are kept constant across models. Deterministic or near-deterministic decoding is used to reduce variability between experimental runs. The generated text is used exclusively for retrieval and is not presented as a recommendation to the user.

The embedding stage uses `intfloat/e5-small-v2`, which produces 384-dimensional dense representations. All candidate-item embeddings are computed once and stored before evaluation. Query-time processing therefore consists of hypothetical-description generation, embedding, cosine similarity computation, and Top- K retrieval.

D. Relevance Assessment

A graded relevance protocol is used to establish evaluation ground truth. Candidate items are assigned one of four relevance levels:

$$r(q, d_i) \in \{0, 1, 2, 3\}, \quad (4)$$

corresponding to non-relevant, weakly relevant, relevant, and highly relevant recommendations, respectively.

To avoid exhaustive annotation of the full candidate corpus, a pooling strategy is adopted. For each learner profile, the Top- K_p candidates returned by all evaluated retrieval methods are combined and duplicates are removed. The resulting candidate pool is evaluated independently by domain experts without revealing which retrieval method selected each item.

Inter-rater agreement is reported using [Krippendorff’s α / weighted Cohen’s κ], depending on the final number of annotators.

E. Evaluation Metrics

Recommendation effectiveness is evaluated using Precision@ K , Recall@ K , Mean Reciprocal Rank (MRR), and normalized Discounted Cumulative Gain (nDCG@ K). nDCG is used as the primary ranking metric because it accounts for both the position and graded relevance of retrieved items.

For a ranked list of length K , DCG is computed as

$$DCG@K = \sum_{i=1}^K \frac{2^{r_i} - 1}{\log_2(i + 1)}, \quad (5)$$

and normalized as

$$nDCG@K = \frac{DCG@K}{IDCG@K}. \quad (6)$$

In addition to recommendation effectiveness, computational efficiency is measured using end-to-end query latency, generation latency, generation throughput, and peak memory consumption.

F. Implementation and Hardware

All experiments are executed locally on an Apple M4 Pro system with 48 GB of unified memory. The embedding model is executed locally and all candidate representations are pre-computed. The evaluated LLMs are deployed locally using the same inference framework and quantization configuration where applicable.

For each generative model, latency is measured over XX repeated queries after an initial warm-up period. The same profile set, prompt template, generation length, and retrieval corpus are used for all configurations to ensure a consistent comparison.

V. RESULTS AND DISCUSSION

A. Recommendation Effectiveness

Table III reports the recommendation performance obtained across both educational tasks. The comparison between Direct-E5 and the HyDE-based configurations directly evaluates whether LLM-mediated semantic alignment improves recommendation quality over direct embedding similarity.

Compared with direct embedding-based retrieval, HyDE-Small [improved/reduced] nDCG@5 by XX% for student-internship matching and XX% for educational-resource recommendation. The corresponding changes in MRR were XX% and XX%, respectively. These results [support/do not support] the hypothesis that transforming learner-side information toward the semantic form of the target representation improves educational recommendation.

The comparison between the two tasks further indicates that [RESULT]. This suggests that [INTERPRETATION REGARDING GENERALIZATION].

B. Effect of Generative Model Capacity

Table IV examines whether increased generative-model capacity translates into higher recommendation effectiveness.

Increasing model capacity from the small to the medium configuration resulted in a [XX%] change in nDCG@5, while query latency increased by [XX%] and peak memory consumption by [XX%]. [If applicable: The limited effectiveness gain suggests that high-capacity generative reasoning is not required for the semantic transformation performed by the proposed framework.]

The cross-model experiment produced [RESULT], indicating that [INTERPRETATION REGARDING MODEL-FAMILY DEPENDENCE].

C. Ablation Analysis

To isolate the contribution of the generative alignment stage, the proposed method is compared with direct dense retrieval while keeping the embedding model, candidate corpus, and similarity function unchanged. Consequently, the only difference between Direct-E5 and HyDE-based retrieval is the transformation

$$q \rightarrow G(q) \quad (7)$$

performed before embedding.

The comparison with generic LLM rewriting is included to determine whether performance gains originate specifically from transforming the learner representation toward the target-item perspective rather than from simply expanding or reformulating the input text.

D. Efficiency on Consumer Hardware

The deployment experiment evaluates whether the semantic-alignment stage can be executed using locally hosted language models without dedicated server-class GPU infrastructure. Table VI reports generation latency, throughput, and memory usage measured on the Apple M4 Pro platform.

The results reveal a [RESULT] trade-off between recommendation effectiveness and computational requirements. In particular, [MODEL] achieved XX% of the nDCG@5 obtained by [LARGER MODEL] while requiring XX% lower latency and XX% less memory, indicating [INTERPRETATION].

VI. CONCLUSION

This work investigated LLM-mediated semantic alignment for personalized educational recommendation. Rather than directly matching heterogeneous learner and candidate-item representations, a HyDE-based intermediate generation stage was used to transform learner information toward the semantic form of the retrieval target before dense embedding and similarity-based ranking.

The framework was evaluated across two complementary higher-education recommendation tasks and compared with lexical retrieval, direct dense retrieval, and alternative generative configurations. Experimental results showed that [MAIN RESULT]. In particular, [BEST CONFIGURATION] achieved [XX] nDCG@5 compared with [XX] for direct embedding-based retrieval, indicating [MAIN INTERPRETATION].

The evaluation of locally deployable language models further showed that [EFFICIENCY RESULT]. These findings suggest that [SMALL/MEDIUM] language models can provide [comparable/sufficient] semantic alignment without requiring server-scale generative models, supporting deployment on consumer hardware and in educational environments with constrained computational infrastructure.

Future work will investigate richer learner representations, additional educational recommendation domains, longitudinal user feedback, and adaptive combinations of semantic retrieval and constrained reranking. Further evaluation on larger and more diverse learner populations will also be required to assess the generalizability of the proposed approach.

TABLE III
RECOMMENDATION EFFECTIVENESS ACROSS THE TWO EDUCATIONAL TASKS. BEST RESULTS ARE SHOWN IN BOLD.

Method	Student–Internship				Educational Resources			
	P@5	R@5	MRR	nDCG@5	P@5	R@5	MRR	nDCG@5
BM25	XX	XX	XX	XX	XX	XX	XX	XX
Direct-E5	XX	XX	XX	XX	XX	XX	XX	XX
HyDE-Small	XX	XX	XX	XX	XX	XX	XX	XX
HyDE-Medium	XX	XX	XX	XX	XX	XX	XX	XX
HyDE-CrossModel	XX	XX	XX	XX	XX	XX	XX	XX

TABLE IV
EFFECT OF LLM CONFIGURATION ON RETRIEVAL EFFECTIVENESS AND EFFICIENCY.

Model	nDCG@5	MRR	Latency	Memory
Small LLM	XX	XX	XX	XX
Medium LLM	XX	XX	XX	XX
Cross-model LLM	XX	XX	XX	XX

TABLE V
ABLATION OF THE SEMANTIC-ALIGNMENT STAGE.

Configuration	MRR	nDCG@5
Original learner text + E5	XX	XX
LLM rewrite + E5	XX	XX
HyDE representation + E5	XX	XX

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TABLE VI
LOCAL INFERENCE EFFICIENCY ON APPLE M4 PRO WITH 48 GB UNIFIED MEMORY.

Model	Latency (s)	Tokens/s	Peak Memory (GB)
Small LLM	XX	XX	XX
Medium LLM	XX	XX	XX
Cross-model LLM	XX	XX	XX